Introduction to Economics

Class 13 — Monopoly; oligopoly and the prisoner's dilemma

Pavel Kocourek

Universitat de Barcelona · Group A6

9 November 2026

Plan for today

- Monopoly: the price and quantity a single seller picks
- What a monopoly costs the rest of us
- A price for each buyer; what governments do
- Oligopoly: two sellers and a deal between them
- More sellers; when a deal can hold
- Practice: today's moves; a tax, a tariff, an external cost

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Block 4's third class, in two halves. Last Monday ended with a mountain's only cable car, a seller that chooses its own price, and left open which price it picks. The first half answers that question and measures what the answer costs everyone else; then two sellers share one market, make a deal and feel the pull to break it, and a real train line that went from one company to three shows what more sellers do to a price. The second half is practice: today's two moves as multiple-choice items, then three markets drawn on a grid, a tax, a tariff and an external cost, each worked on the board the way you would work it on paper.

Slide 2 Plan for today

The class at a glance, in two halves. Before the break, the ideas: the cable car's demand table, why each extra ride adds less to its revenue than its price, the number of rides and the price it picks, what that choice costs the riders who stay at home, a price for each rider and what governments do; then two farms that share a bay, a deal between them, why each is tempted to break it and what happens when both do, why more sellers push a price toward cost, and when a deal can hold anyway.

After the break, practice: two multiple-choice items on today's two moves, then three graph problems, a tax, a tariff and an external cost, in the form of the practice exam on the course page: you try each alone, then it is solved on the board the way you would solve it on paper.

Madrid–Barcelona by fast train: one company, then three

Revealed in class

Before class: Ouigo (May 2021) and Iryo (November 2022) started running rival trains on this line. Guess what had happened by 2025 to the average fare and to the number of passengers — and why.

Source given in class.

Slide 3 Madrid–Barcelona by fast train: one company, then three

A real market where the number of sellers changed: on the high-speed line between Madrid and Barcelona one company, Renfe, sold every ticket until Ouigo arrived in May 2021 and Iryo in November 2022. Before class, guess what had happened by 2025, compared with 2019, to the average fare paid (total ticket revenue divided by the number of passengers) and to the number of passengers, and ask why two newcomers would matter when they are only two.

Revealed in class.

From the second hundred on, each adds less to revenue than its price: the price falls for all

highest price for all (euros)	13	12	11	**10**	9	**8**	7	6
rides a day	100	200	300	**400**	500	**600**	700	800
revenue a day (euros)	1,300	2,400	3,300	**4,000**	4,500	**4,800**	4,900	4,800
marginal revenue (euros a ride)	13	11	9	**7**	5	**3**	1	−1

Marginal revenue: what one more ride adds to revenue
(here: per ride, for each extra hundred)

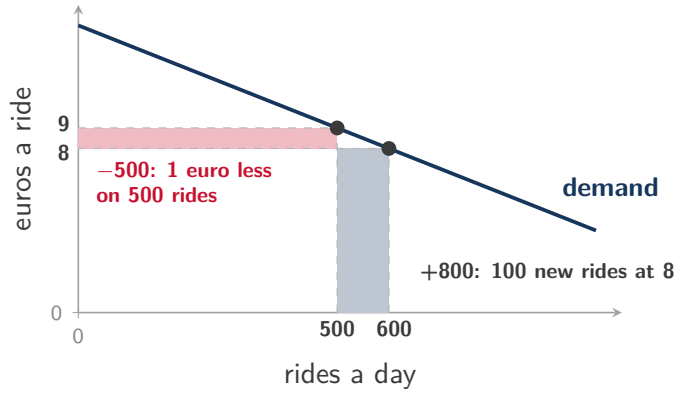
Monopoly: the only seller of a good with no close substitute; it faces the market's demand (class 12, slide 20).

A price taker's extra bag adds just the price: marginal revenue (MR) = P = 4 (class 12, slides 6 and 8).

Slide 4 From the second hundred on, each adds less to revenue than its price: the price falls for all

Back to last Monday's cable car, the only one up its mountain, with the same demand: for each number of rides a day, the highest price at which all of them sell, one price for every rider. Revenue is price times rides; marginal revenue is what one more ride adds to revenue, here measured per ride for each extra hundred. The gap between the top row (price) and the bottom row (marginal revenue) is the whole difference between today and last Monday's print shop: the shop could sell any number of bags at 4 euros, so one more bag added exactly the price; the cable car can sell more only by lowering the price for everyone, so each extra hundred adds less than its price, and past 700 rides it adds less than nothing. The next slide shows where the gap comes from.

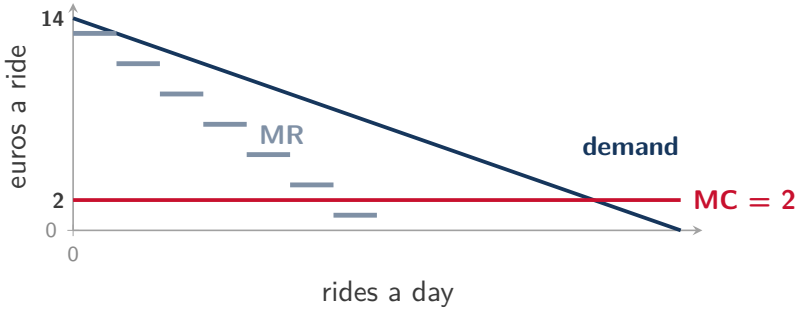
From 500 to 600 rides: 800 more from new riders, 500 less from the old ones



Marginal revenue: 800 − 500 = 300 for 100 rides — 3 a ride, below the price of 8.

Which rides does the cable car sell?

Each extra ride costs 2 euros (energy, wear); fixed costs are paid whatever it sells.



Before class: which is the last hundred whose marginal revenue beats the 2-euro cost? What price goes with it?

Marginal cost (MC): what one more ride adds to total cost (class 11, slide 10). A price taker produces while MC is below the price (class 12, slide 8).

Slide 5 From 500 to 600 rides: 800 more from new riders, 500 less from the old ones

The gap in slide 4's table, drawn. Moving down the demand curve from 500 rides at 9 to 600 at 8 has two effects that pull against each other: the hundred new riders pay 8 each, but the five hundred who were already riding at 9 now pay a euro less each. Marginal revenue is what is left after the second effect, which is why it sits below the price. A price taker never feels the second effect, because it never has to lower its price to sell more; a monopoly always does. The slide assumes one price for all riders; what the cable car could do if it could charge the old riders 9 and the new ones 8 comes on slide 8.

Slide 6 Which rides does the cable car sell?

The monopoly's choice. New today is the cost side: each extra ride costs 2 euros, while the cables and cabins cost the same whatever it sells, so they cannot change the best number of rides. The steps are slide 4's marginal revenue, one flat step for each hundred rides. Before class, run along the steps and ask which is the last hundred that adds more to revenue than to cost, and then which price goes with that number of rides on the demand curve.

Revealed in class.

What does the cable car's choice cost?

Revealed in class

Before class: at the cable car's price, some riders who value a ride at more than its 2-euro cost do not ride. How many rides would be sold at a price of 2? Who loses what? (The demand line: price $=$ 14 $-$ rides $\div$ 100.)

Slide 7 What does the cable car's choice cost?

The cost of the cable car's choice to riders and cable car together. At its price, some riders who value a ride above its 2-euro cost do not ride. Before class, ask how many rides would be sold at a price of 2 (the demand line: price $=$ 14 minus rides divided by 100), who keeps what at the cable car's price, and what happens to the value of the rides that are not sold.

Revealed in class.

A different price for each rider?

Revealed in class

Price discrimination: selling the same good at different prices to different buyers.
Before class: if every rider paid exactly what the ride is worth to them, how many rides would the cable car sell, and who would gain the surplus?

Slide 8 A different price for each rider?

Price discrimination means selling the same good at different prices to different buyers. Before class, suppose every rider paid exactly what the ride is worth to them: ask how many rides the cable car would then sell, compared with one price for all, and who would end up with the surplus.

Revealed in class.

Four answers to a monopoly: competition law, a price cap, public ownership — or nothing

- **Competition law:** stop mergers, fine price-fixing
- **Regulation:** cap the price a natural monopoly may charge
- **Public ownership:** the state runs it
- **Doing nothing:** when every cure costs more

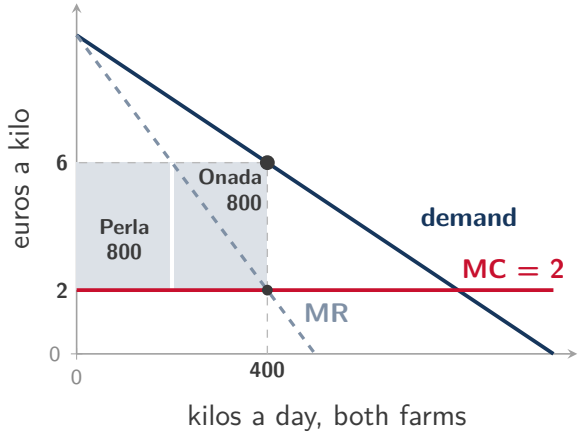
Natural monopoly: one firm serves the whole market most cheaply (class 12, slide 21).

In Spain: the CNMC (Comisión Nacional de los Mercados y la Competencia). Reading Ch. 11, pp. 258–261.

Slide 9 Four answers to a monopoly: competition law, a price cap, public ownership — or nothing

Four things a government can do about a monopoly, each with its own cost. Competition law blocks mergers that would create one and fines firms that fix prices; in Spain the body in charge is the CNMC, the same authority that wrote the train report of slide 3. Regulation caps the price a natural monopoly may charge, such as class 12's electricity grid; the cap cannot simply be set at marginal cost, because a natural monopoly's average cost lies above its marginal cost (the fixed cost is spread over the units), so at that price it would lose money; regulators allow a price near average cost, or a cap that falls year by year to keep the pressure on costs. Public ownership removes the profit motive, and with it much of the pressure to keep costs down. Doing nothing is right when the loss is small and every cure would cost more. Any news story about a regulator or a competition authority fits one of these four boxes.

Two mussel farms share one bay: acting as one monopoly, they would sell 400 kilos at 6 euros



Perla and Onada: the only two farms; each kilo costs 2 to farm.

Oligopoly: a market with a few sellers of similar goods

Cartel: sellers who agree to act as one monopoly: here 200 kilos each

Slide 10 Two mussel farms share one bay: acting as one monopoly, they would sell 400 kilos at 6 euros

From one seller to two. Perla and Onada are the only farms in the bay, so together they face the market's demand, and each kilo costs either of them 2 euros to farm. A market with a few sellers is an oligopoly, and its mark is that each seller's profit depends on what the other does. If the two acted as one firm they would use the cable car's rule: marginal revenue equals marginal cost at 400 kilos, the price read off demand is 6, and the 1,600 a day of profit splits into 800 each when each farms 200 kilos. An agreement to do this is collusion, and the group is a cartel. It pays them because together they are a monopoly, with a price far above cost: two sellers do not have to compete the price down to cost if they can agree. The next slide asks whether the agreement holds.

Keep the deal or sell more? Each farm's profit a day

Each farm sells 200 kilos (the deal) or 250.
Price: 400 kilos → 6.00 · 450 → 5.50 · 500 → 5.00.

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

$$\text{profit} = (\text{price} - 2) \times \text{own kilos}$$

Slide 11 Keep the deal or sell more? Each farm's profit a day

The cartel in a payoff table. Each farm picks 200 kilos (the deal) or 250; the price depends on the total, and each cell shows both farms' profit a day, Perla's first, as (price minus 2) times own kilos. Before class, check one cell against the setup line, then take Perla's point of view: for each thing Onada might do, compare Perla's two numbers and ask what Perla would do.

Revealed in class.

Where do the two farms end up?

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

Dominant strategy: best whatever the other does

Nash equilibrium: each does its best, given what the other does

Prisoner's dilemma: each does its best, and both are worse off than if both had cooperated

Slide 12 Where do the two farms end up?

The same table with three terms to learn: dominant strategy, Nash equilibrium, prisoner's dilemma. Before class, for each choice the other farm might make, circle each farm's best reply; look for a cell where both numbers are circled, and compare it with the deal.

Revealed in class.

One seller, two, many: where does the price go?

Revealed in class

Before class: as one monopoly the two farms sell 400 kilos at 6 euros; each kilo costs 2. Where does the price go when each farm does its best for itself, and when many farms share the bay? Why?

Slide 13 One seller, two, many: where does the price go?

Back to slide 3's question, with the two farms. As one monopoly they sell 400 kilos at 6 euros, and each kilo costs 2. Before class, ask where the price goes when each farm does its best for itself, and where it goes when many farms share the bay; then ask why a seller with a small share of the market is keener to sell more than a monopoly is.

Revealed in class.

Meeting every day: can the deal hold?

Revealed in class

Before class: Perla breaks the deal on day 1, and from day 2 both sell 250 kilos every day. How much does Perla gain on day 1, and lose on each day after? Tit-for-tat: start with the deal, then do what the other did yesterday.

Slide 14 Meeting every day: can the deal hold?

The two farms do not play once but meet every morning at the market. Before class, suppose Perla breaks the deal on day 1 and from day 2 both sell 250 kilos every day: work out from slide 11's table how much Perla gains on day 1 and loses on each day after, and how many days it takes for the gain to be used up. Tit-for-tat means: start with the deal, then do what the other did yesterday.

Revealed in class.

A monopoly sells while marginal revenue beats marginal cost and reads its price off the demand curve.

Its price is above marginal cost: rides worth more than their cost go unsold.

slide 15 of 25

Slide 15 In one line

The first half in four lines, to read slowly: the first is slides 4 to 6, the second slide 7. The third line (slides 10 to 12) and the fourth, which answers slide 3's question, are revealed in class.

Break

Ten-minute break

slide 16 of 25

Slide 16 Break

Ten minutes. Paper copies of Problem set 3 are handed out now: your copy number is the same as on Campus Virtual. After the break: practice, two multiple-choice items and three graph problems.

Multiple choice, two minutes alone: the monopoly's price

A1 [4 points]

One firm owns the town's only fibre network. Each extra household it connects costs it 10 euros a month. The table gives the highest price at which it can connect each number of households, one price for all. Which price gives the firm the largest profit?

price (euros a month)	60	50	40	30	20	10
households (hundreds)	1	2	3	4	5	6

A 20 euros a month
B 40 euros a month
C 10 euros a month
D 30 euros a month

Answer A1

(A) (B) (C) (D)

Slide 17 Multiple choice, two minutes alone: the monopoly's price

The move of slides 4 to 6 as a multiple-choice item, solved in class on the board the way you would on paper. Try it alone before class: write the revenue of each row, then what each extra hundred adds per household, decide where to stop against the 10-euro cost, and read the price from the right row. Then ask what mistake each of the other three options would be.

Revealed in class.

Multiple choice, two minutes alone: open on Sundays?

A2 [4 points]

A town's only two supermarkets, Fresc and Rodó, each choose whether to open on Sundays. Profit in thousand euros a month, Fresc's first:

		Rodó closed	**Rodó** open
Fresc	closed	Fresc 12, Rodó 10	Fresc 8, Rodó 13
	open	Fresc 15, Rodó 6	Fresc 9, Rodó 8

Which outcome is the Nash equilibrium?

A Fresc: closed; Rodó: closed.
B Fresc: open; Rodó: closed.
C Fresc: closed; Rodó: open.
D Fresc: open; Rodó: open.

Answer A2

(A) (B) (C) (D)

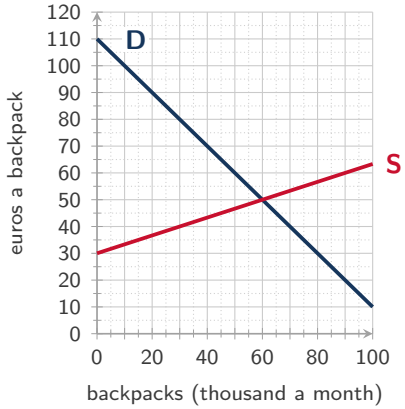
Slide 18 Multiple choice, two minutes alone: open on Sundays?

Slide 12's method on a new table, as a multiple-choice item: two supermarkets, each open on Sundays or closed, profits in thousand euros a month. Try it alone before class: for each choice of the other shop, circle a shop's own best reply, comparing only its own two numbers; then look for the cell where both numbers are circled, and ask what each other option would mean.

Revealed in class.

A tax on backpacks: the line and the prices — three minutes alone

The grid shows the market for backpacks: demand D and supply S. The state puts a tax of 40 euros on every backpack sold; the sellers pay it to the state.



[5 points; drawing 2; boxes as marked]

(a) Draw the supply curve with the tax: at every quantity 40 euros above S. Label it S + tax.

(b) With the tax (quantity in thousand, prices in euros):

Quantity sold [1] | Buyers pay (euros a backpack) [1]

Sellers keep (euros a backpack) [1]

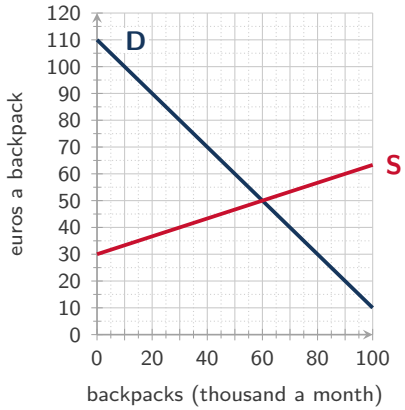
Draw on the grid and label every line you draw; numbers in the boxes, working beside them.

Slide 19 A tax on backpacks: the line and the prices — three minutes alone

The first graph problem, in the form of the practice exam on the course page, solved in class on the board step by step. Try it alone before class on a sketch of the grid: read the scale, draw the supply curve with the tax and label it, then fill the three boxes with the working beside them.

Revealed in class.

The tax's areas: numbers, and letters inside — five minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax of 40 euros a backpack, in thousand euros a month: CS, PS, T and DWL (below). Write each one's letters inside its area.

CS [1.5] | PS [1.5] | T [1.5] | DWL [1.5]

(d) Who bears more of the tax? Circle one. [1]

Buyers | Sellers

(e) Same tax, but the buyers pay it to the state: which line would you draw? Would any box change?

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. An area's letters go inside it (a small area: beside it, with an arrow). A triangle is ½ × base × height.

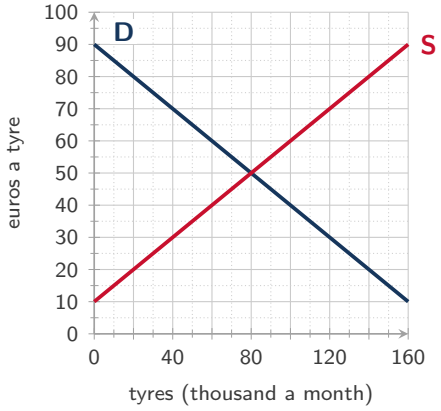
Slide 20 The tax's areas: numbers, and letters inside — five minutes alone

The same market: the four areas with their letters, who bears more of the tax, and what changes if the buyers pay it instead. Try it alone before class on your sketch from the previous slide: shade each area, write its letters inside it, and compute each one, ½ × base × height for a triangle.

Revealed in class.

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1] Consumption [1] Imports [0.5]

(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5] Production [0.5]

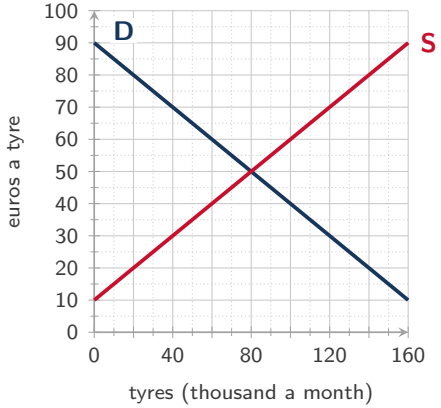
Consumption [0.5] Imports [0.5]

Slide 21 Tyres: opening to trade, then a tariff — four minutes alone

A second market in the practice exam's form: a country opens to trade at a world price, then puts a tariff on imports; solved in class on the board. Try it alone before class: read the scale (it differs from the first grid), draw each price as a flat line with its label, and decide which curve gives production and which consumption before you fill the boxes.

Revealed in class.

The tariff's areas: T and two DWL triangles — three minutes alone



[5 points; letters 1; boxes as marked]

(c) With the tariff (world price 30, tariff 10), in thousand euros a month: the tariff revenue (T) and the deadweight loss (DWL, both triangles together). Write T inside its area, and DWL beside each of its two triangles, with an arrow.

T [2]

DWL (total) [2]

Thousand tyres × euros = thousand euros. A triangle is ½ × base × height.

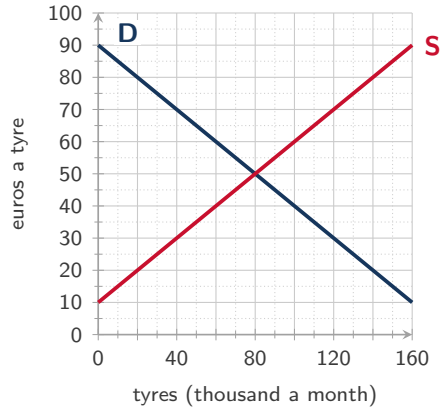
Slide 22 The tariff's areas: T and two DWL triangles — three minutes alone

The same market with the tariff: its revenue and its deadweight loss, each with its letters placed as the practice exam asks. Try it alone before class on your sketch from the previous slide: find the rectangle and the two triangles first, then compute them in thousand euros.

Revealed in class.

The same tyres with smoke: an external cost — four minutes alone

The tyre market without trade, D and S as before. Making each tyre also sends out smoke that harms people nearby: 30 euros of harm per tyre, which buyers and sellers do not count.



[10 points; drawing 4; boxes as marked]

(a) Draw the social-cost curve (SC: the sellers' cost plus the harm to others); label it SC.

(b) Market quantity [1] Efficient quantity [1.5]

(c) The deadweight loss (DWL) at the market quantity, thousand euros a month; label its area DWL.

(d) The tax per tyre that brings the market to the efficient quantity.

(c) DWL [2] (d) Tax (euros a tyre) [1.5]

Slide 23 The same tyres with smoke: an external cost — four minutes alone

The tyre market again, without trade and with an external cost, in the practice exam's form; solved in class on the board. Try it alone before class: draw the social-cost curve and label it, mark the market quantity and the efficient quantity, shade the deadweight-loss triangle with its letters inside, and say which tax per tyre would fix it.

Revealed in class.

Hoodies: from costs to a choice — four minutes alone (optional)

[8 points; boxes as marked]

A workshop sews hoodies and sells them at the market price. Its total cost, euros a week:

hoodies a week	0	25	50	75	100
total cost (euros)	400	700	900	1,300	1,900
marginal cost, MC (euros a hoodie)					

(a) Fill in the marginal cost (MC) of each extra 25 hoodies, per hoodie. [2]

(b) At 50 hoodies: the average variable cost (AVC)? The average total cost (ATC)? [2]

(c) The price is 20 euros a hoodie. How many hoodies? Profit? [2]

(d) The price falls to 14. Produce or shut down? Why? [2]

Variable cost = total cost − fixed cost (class 11, slides 9–11).
A price taker produces while MC < price (class 12, slide 8).

Slide 24 Hoodies: from costs to a choice — four minutes alone (optional)

An optional problem in the firm's form: a cost table, the marginal cost of each batch, the two averages, the best quantity at a price and whether to stay open when the price falls. Try it alone before class: find the fixed cost in the table first, then ask which average leaves it out and which rule the shutdown decision uses.

Revealed in class.

Before Thursday 12 November

- Thursday 12 November: monopolistic competition; problems on Blocks 3–4 — read Mankiw & Taylor Ch. 12, pp. 267–276
- Problem set 3: paper copy handed out today; bring your worked solutions and your copy on Thursday (not handed in)
- Practise in Sextant: the units the course page lists



pkocourek@ub.edu · pkocourek.com/teaching/intro-economics · pkocourek.com/sextant/?c=intro-econ

Slide 25 Before Thursday 12 November

Thursday 12 November: monopolistic competition, many sellers each with its own slightly different product, and problems on Blocks 3 and 4. Reading: Mankiw & Taylor chapter 12, pages 267 to 276. Problem set 3, on Block 4, is not handed in: paper copies of Problem set 3 were handed out today, and your copy number is the same as on Campus Virtual; work it on your own paper and bring your worked solutions, with your copy of the set, to class on Thursday 12 November. Its problems 4 and 5 use today's two first-half moves, the demand table and the two-by-two game. For the graphs, the practice exam on the course page: do it in one sitting; its sample solution follows there. Practise in Sextant with the units the course page lists.

pkocourek.com/teaching/intro-economics · pkocourek@ub.edu · pkocourek.com/sextant/?c=intro-econ